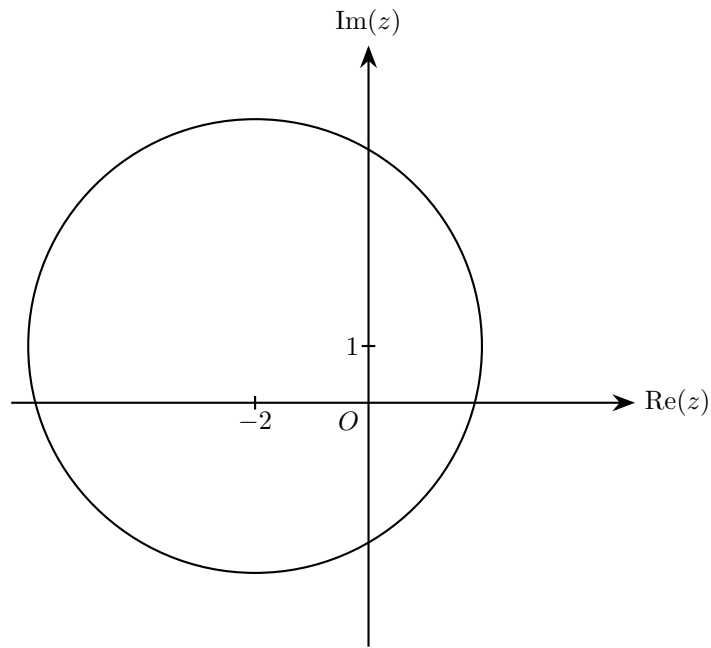


## Questions

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1. The Argand diagram shows a circle of radius 4. The centre of the circle is the point which represents the complex number  $-2 + i$ .

- (a) Use set notation to define the locus of complex numbers,  $z$ , represented by points which lie on the circle. [2]

The locus  $L$  is defined by

$$L = \left\{ z \in \mathbb{C} \mid |z - (-2 + i)| = |z + 1 - 3i| \right\}$$

- (b) On a copy of the Argand diagram, sketch and label the locus  $L$ . [2]

You are given that the locus

$$\left\{ z \in \mathbb{C} \mid \arg(z + 1 - 2i) = \frac{2\pi}{3}, \operatorname{Im}(z) = 5 \right\}$$

contains only one number.

- (c) Find this number. [2]

**2.** In an Argand diagram, the point  $P$  representing the complex number  $w$  lies on the locus defined by

$$\left\{ z \in \mathbb{C} \mid \arg(z - 2 + i) = \frac{\pi}{6} \right\}$$

You are given that  $\operatorname{Im}(w) = 3$ .

**(a)** Find  $w$ .

[2]

The point  $P$  also lies on the locus defined by  $\left\{ z \in \mathbb{C} \mid |z - 2 - 11i| = k \right\}$ , where  $k$  is a constant.

**(b)** Find the complex number represented by the other point of intersection of the loci defined by

$$\left\{ z \in \mathbb{C} \mid |z - 2 - 11i| = k \right\} \quad \text{and} \quad \left\{ z \in \mathbb{C} \mid \arg(z - 2 + i) = \frac{\pi}{6} \right\}$$

[7]

**3.** On an Argand diagram, the locus of points satisfying  $|z - 4 + 3i| = 8$  is a circle.

**(a)** Write down

(i) the coordinates of the centre of this circle [1]

(ii) the radius of this circle [1]

**(b)** On an Argand diagram, shade the set of points defined by

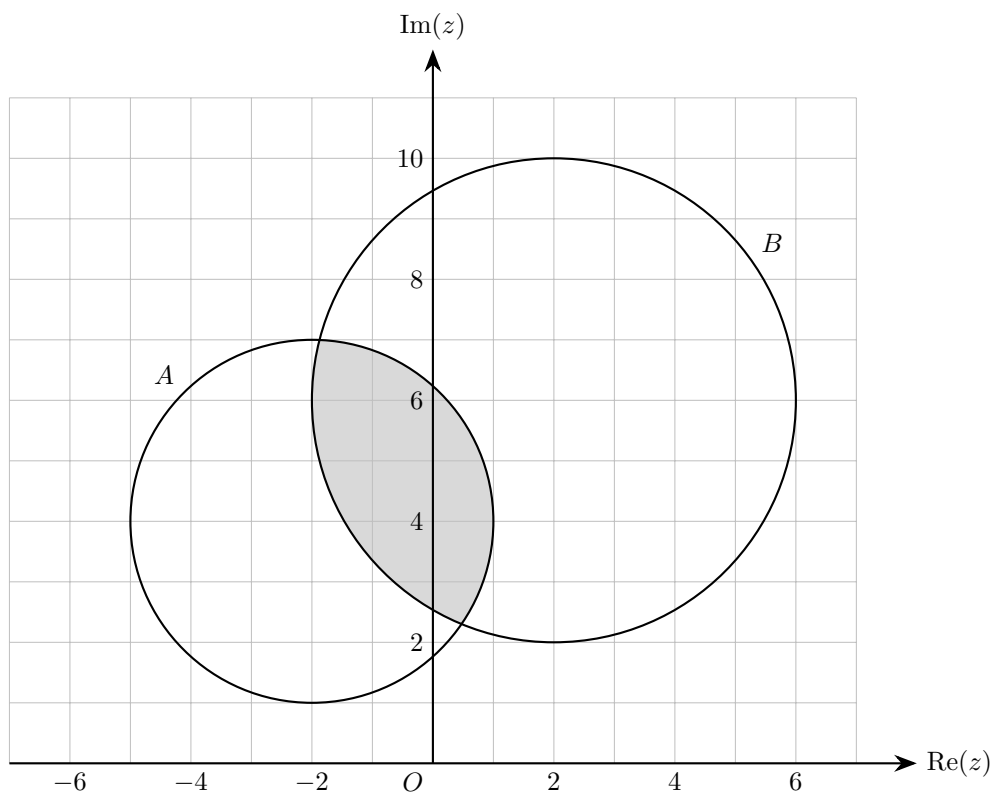
$$|z - 4 + 3i| \leq 8 \quad [1]$$

**(c)** For the set of points defined in part (b), determine the maximum value of  $|z|$ . [3]

The set of points  $A$  is defined by

$$A = \left\{ z \in \mathbb{C} \mid -\frac{\pi}{4} \leq \arg(z - 3 - 6i) \leq \frac{3\pi}{4} \right\} \cap \left\{ z \in \mathbb{C} \mid |z - 4 + 3i| \leq 8 \right\}$$

**(d)** Determine the area of the region defined by  $A$ , giving your answer to 3 significant figures. [4]



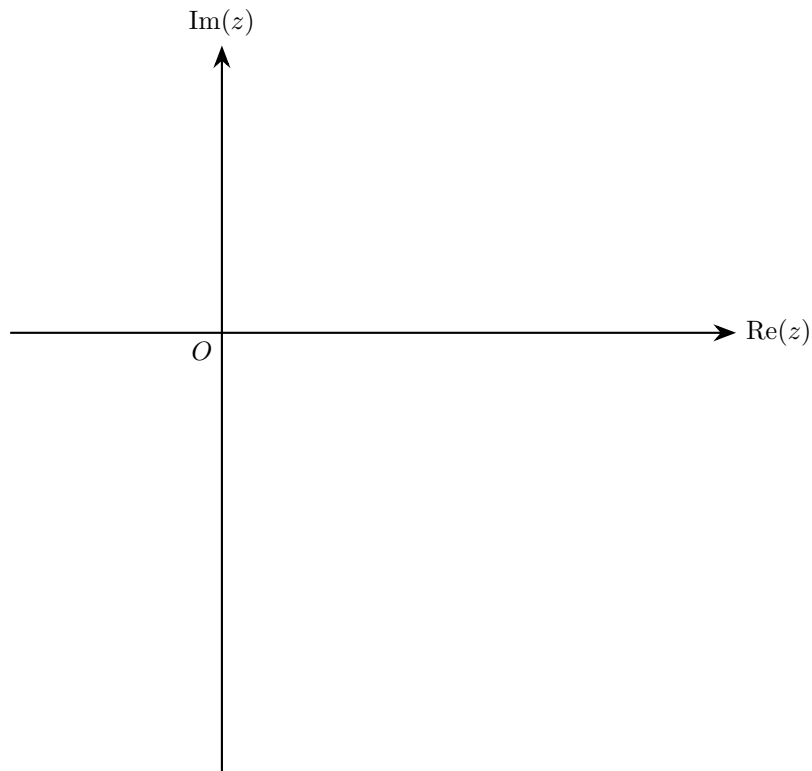
4. (a) The diagram above shows an Argand diagram.

- (i) Write down the equation of the locus represented by the circumference of circle  $A$ . [3]
- (ii) Write down the two inequalities that define the shaded region, not including either circumference. [3]

(b) (i) Draw an Argand diagram to show the region where

$$-\frac{\pi}{6} < \arg(z + 2 - i) < \frac{\pi}{4} \quad [3]$$

- (ii) Determine whether the point  $1 + 5i$  lies within this region. [3]



5. (a) Find the modulus of the complex number  $2 + 2\sqrt{3}i$ , giving your answer as an integer. [2]
- (b) The locus of points,  $L$ , satisfies the equation  $|z - (2 - 2\sqrt{3}i)| = 2$ . [3]
- (i) Sketch the locus  $L$  on the Argand diagram below. [3]
- (ii) The complex number  $w$  lies on  $L$  so that  $-\pi < \arg w \leq \pi$ . Find the greatest possible value of  $\arg w$ , giving your answer in terms of  $\pi$ . [2]
- (c) Solve the equation  $z^4 = 2 + 2\sqrt{3}i$ , giving your answers in the form  $re^{i\theta}$ , where  $r > 0$  and  $-\pi < \theta \leq \pi$ . [5]

6. The point  $P$  represents a complex number  $z$  on an Argand diagram such that

$$|z - (1 + i)| = \frac{1}{2}|z - (4 + i)|$$

- (a) Show that, as  $z$  varies, the locus of  $P$  is a circle, and give the coordinates of the centre and the radius of the circle. [5]

The point  $Q$  represents a complex number  $z$  on an Argand diagram such that

$$|z - (2 + 3i)| = |z + 2 - i|$$

- (b) Sketch, on the same Argand diagram, the locus of  $P$  and the locus of  $Q$  as  $z$  varies. [5]

- (c) On your diagram shade the region which satisfies

$$|z - (1 + i)| \leq \frac{1}{2}|z - (4 + i)| \quad \text{and} \quad |z - (2 + 3i)| \geq |z + 2 - i| \quad [2]$$

7. On an Argand diagram, the point  $P$  representing the complex number  $z$  lies on the locus

$$|z - (2 - i)| = 3$$

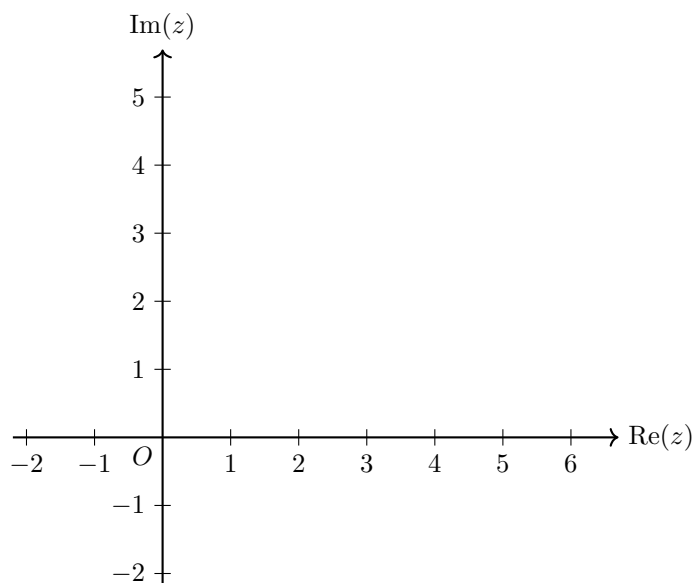
The point  $Q$  representing the complex number  $w$  lies on the locus

$$|w + 1 - 11i| = 3|w + 1 - 3i|$$

Find the exact coordinates of the points common to these two loci.

[7]





8. (a) Sketch, on the Argand diagram above, the locus of points satisfying the equation

$$|z - (3 + i)| = 1 \quad [1]$$

- (b) There is a unique complex number  $w$  that satisfies both

$$|w - (3 + i)| = 1 \quad \text{and} \quad \arg(w) = \alpha$$

where  $\alpha$  is a constant such that  $0 < \alpha < \pi$ .

- (i) Find the value of  $\alpha$ . [2]  
(ii) Find  $w$  and express it in the form  $r(\cos \theta + i \sin \theta)$ . Give each of  $r$  and  $\theta$  to two significant figures. [4]

**9.** The locus of points  $z = x + iy$  that satisfy

$$\arg \left( \frac{z - (2 + i)}{z - (6 + 5i)} \right) = \frac{\pi}{4}$$

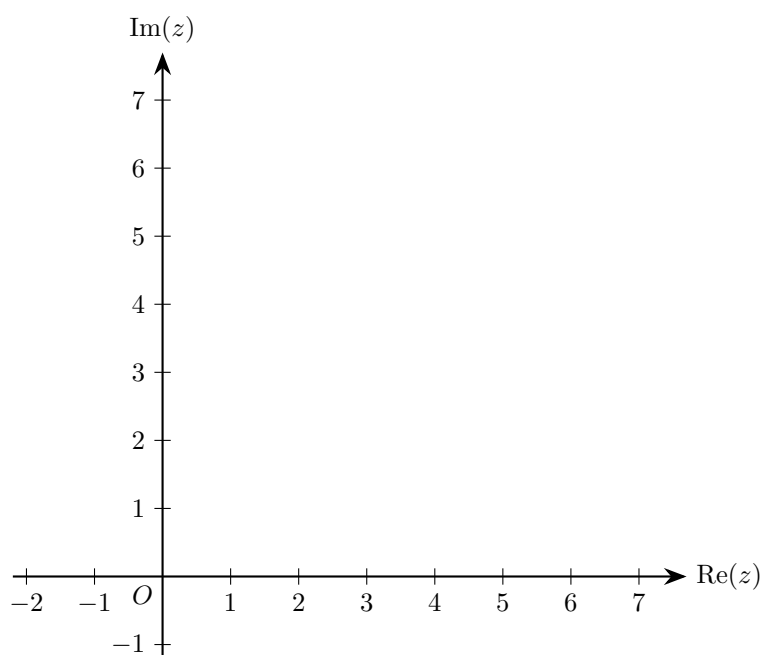
is an arc of a circle  $C$ .

(a) On an Argand diagram sketch the locus of  $z$ . [2]

(b) Explain why the centre of  $C$  lies on the line  $y = 7 - x$ . [1]

(c) Determine the radius of  $C$ . [2]

(d) Determine the coordinates of the centre of  $C$ . [2]



10. (a) Sketch, on the Argand diagram below, the locus of points satisfying the equation

$$|z - (2 + 4i)| = 3 \quad [2]$$

- (b) Sketch, also on the Argand diagram, the locus of points satisfying the equation

$$\arg(z - 1) = \frac{\pi}{4} \quad [1]$$

- (c) For the complex number  $w$ , find the maximum value of  $|w|$  such that

$$|w - (2 + 4i)| \leq 3 \quad \text{and} \quad 0 \leq \arg(w - 1) \leq \frac{\pi}{4} \quad [3]$$